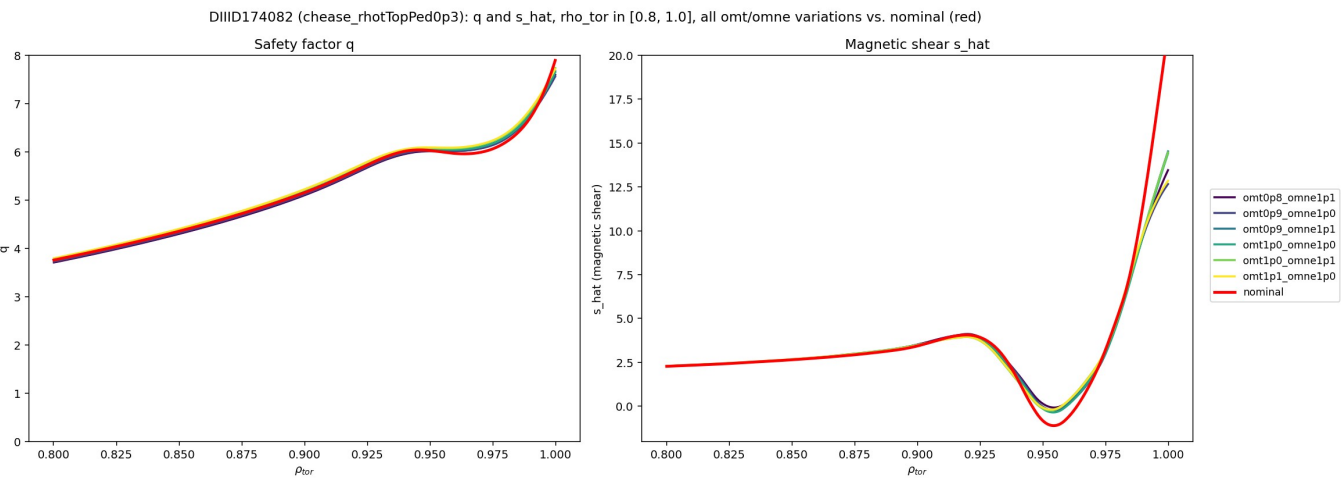


Convergence difficulties in CHEASE can be solved by relaxing the q0 (and Ip) constraint (NCSCAL=4 instead of NCSCAL=1 in CHEASE namelist)

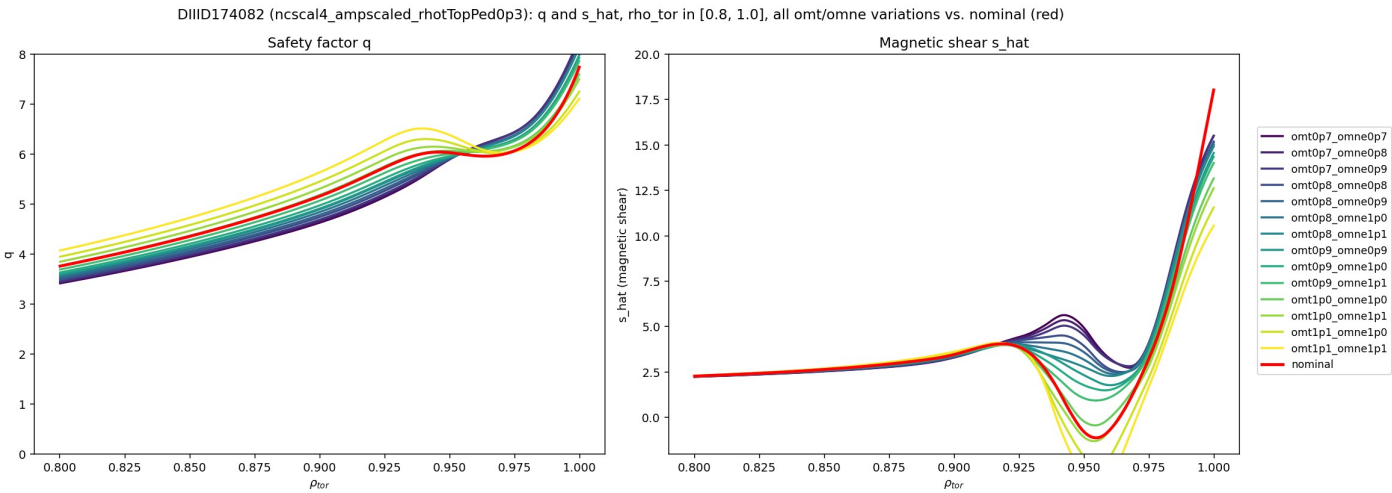
rhotTopPed = 0.3 (large core profile changes)
With Ip constraint (amplitude rescaling in outer loop)
With q0 constraint (NCSCAL = 1)
→ difficult convergence, no expected shear on pressure dependence

No q0 constraint (NCSCAL = 4)
→ good convergence, expected shear on pressure dependence

Likely the preferred solution
If original setup fails.



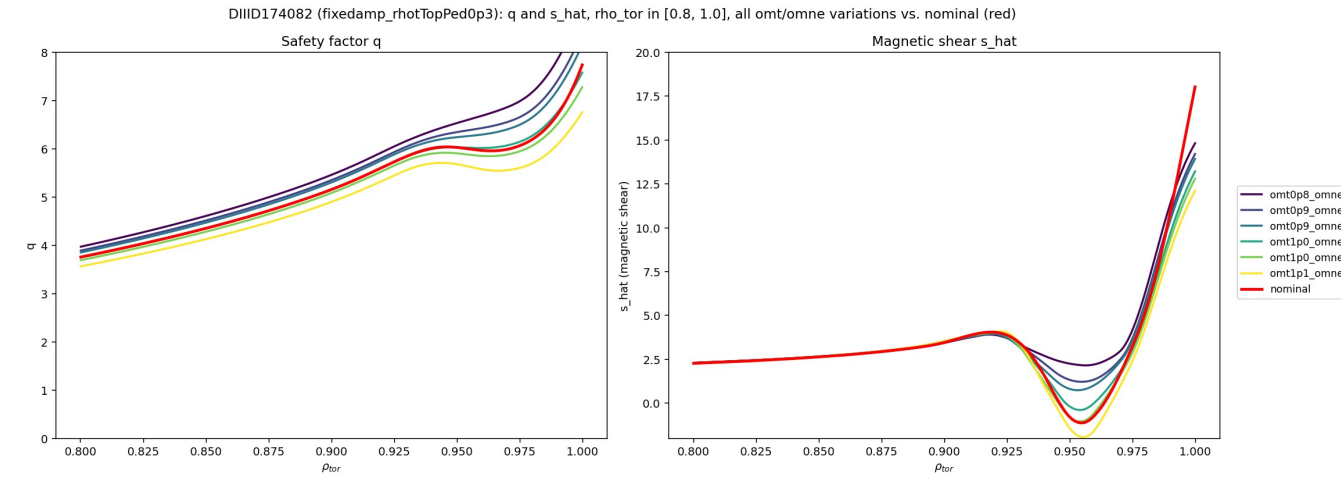
Relax q0



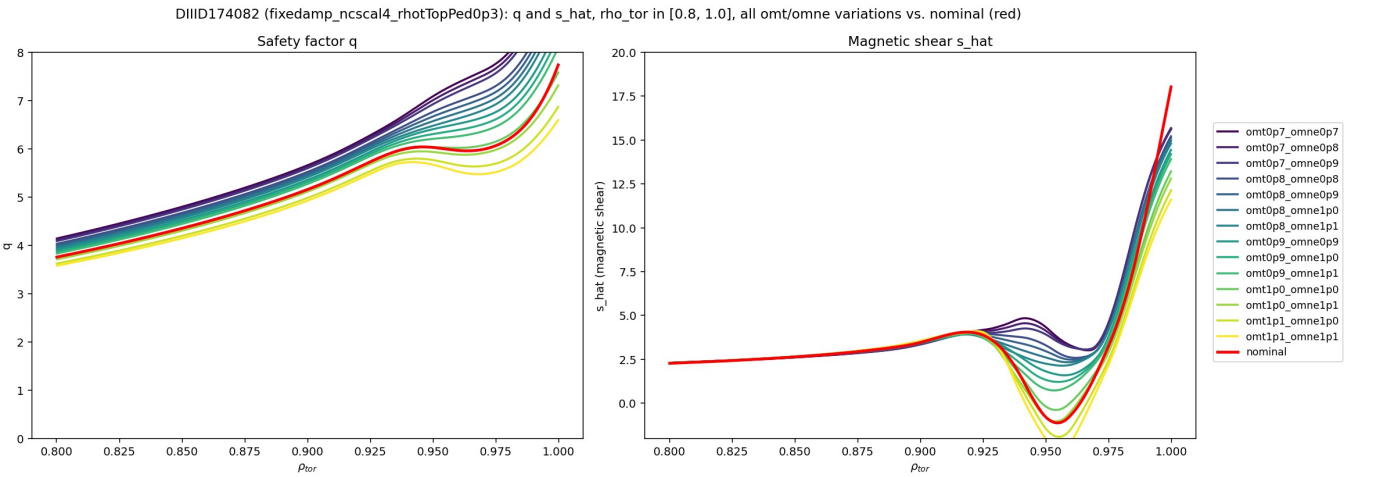
Relax Ip

No Ip constraint (fixed amplitude in python loop)
→ improved, but not good convergence and shear dependence

No q0 constraint and no Ip constraint
→ good convergence, expected shear on pressure dependence
But Ip not preserved

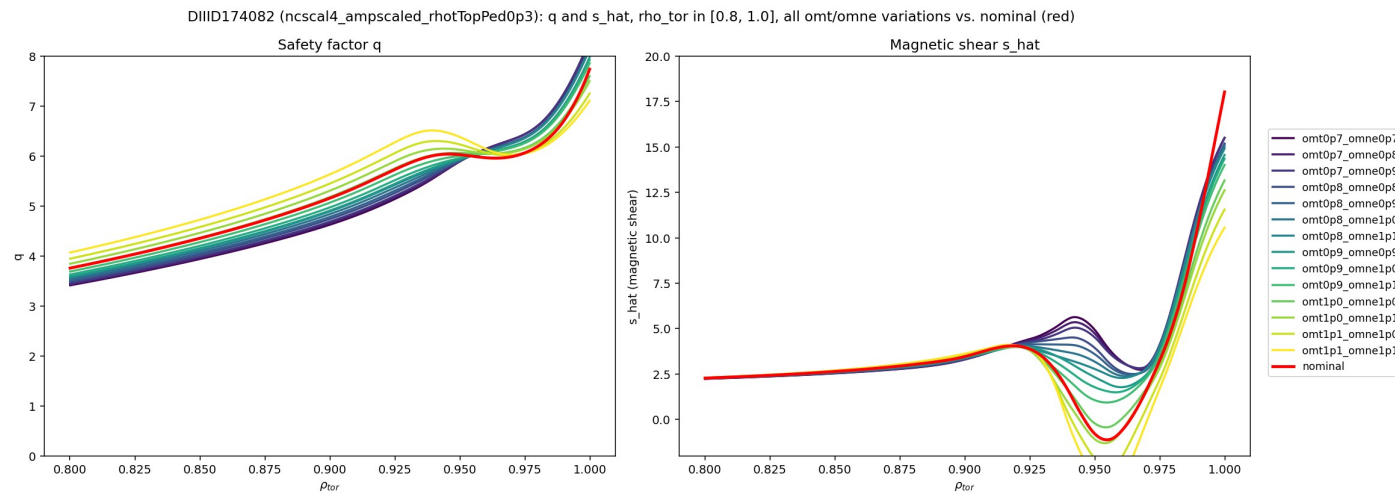


Relax q0

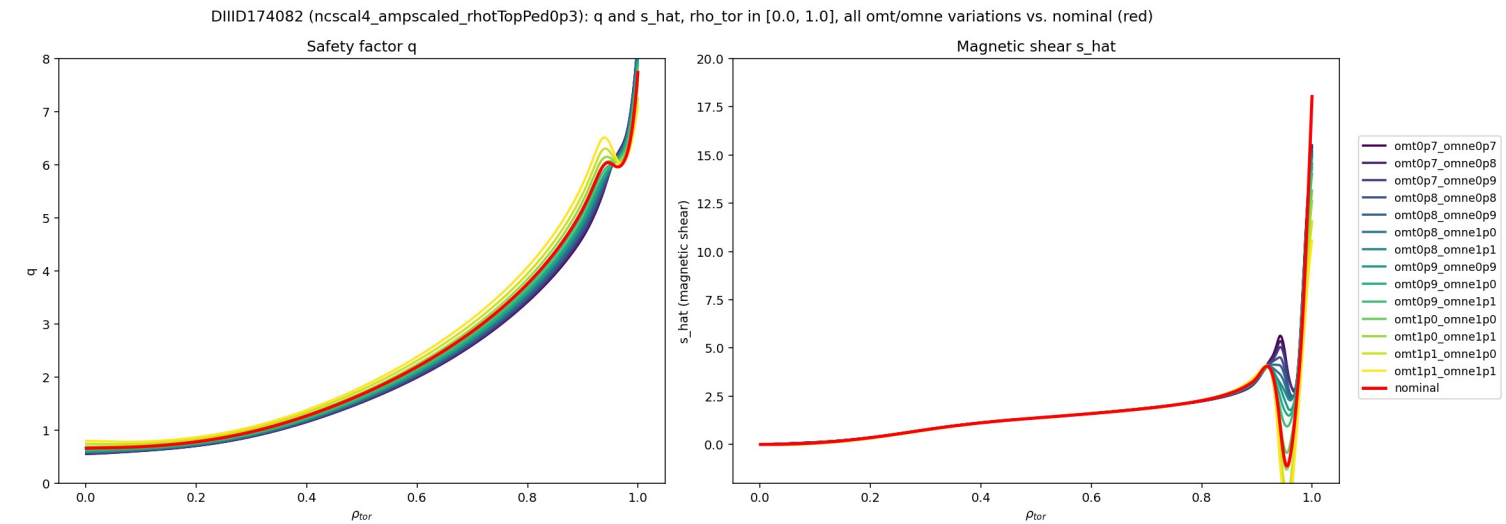


Closer look at NCSCAL =4 option (no q0 constraint)

Pedestal zoom



Full profile

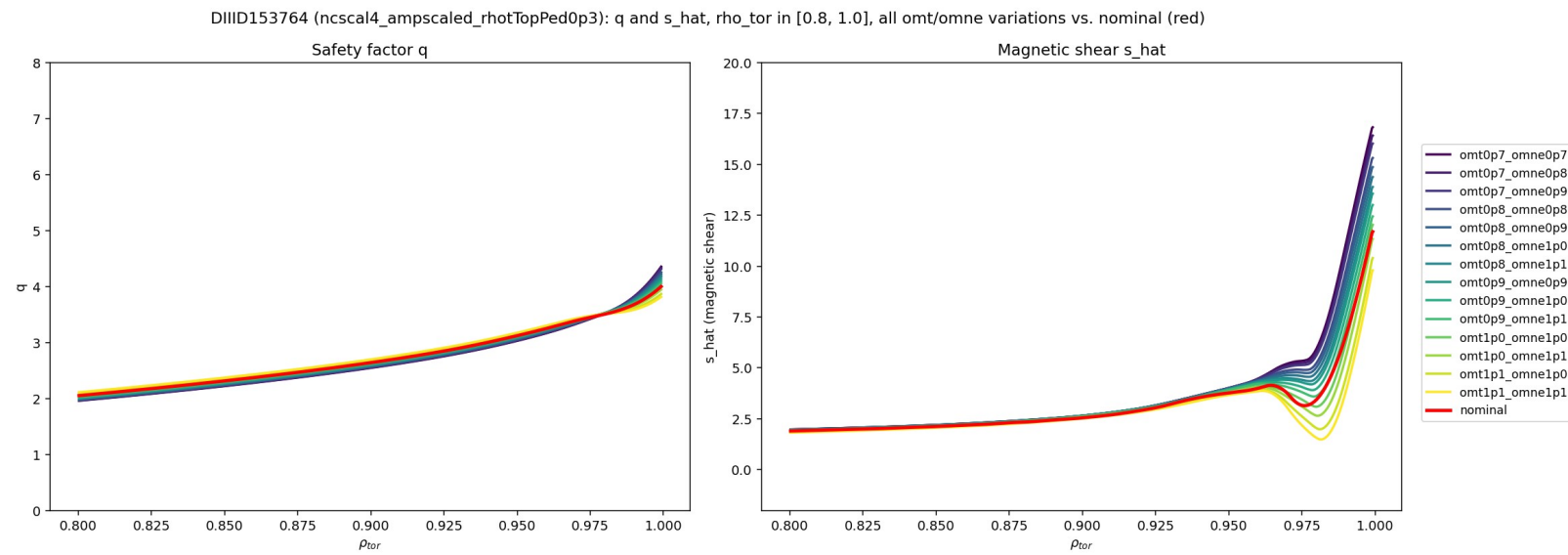


- In example (DIIID174082) the q0 variation without constraint is < 25%.
- Under NCSCAL = 4, q0 is determined from on-axis driven current and not rescaled to match target.
- Reassuring: convergence becomes easier when constraints are relaxed.
- Keeping q0 to nominal value when core pressure profiles change a lot seems genuinely hard to fulfill/in conflict

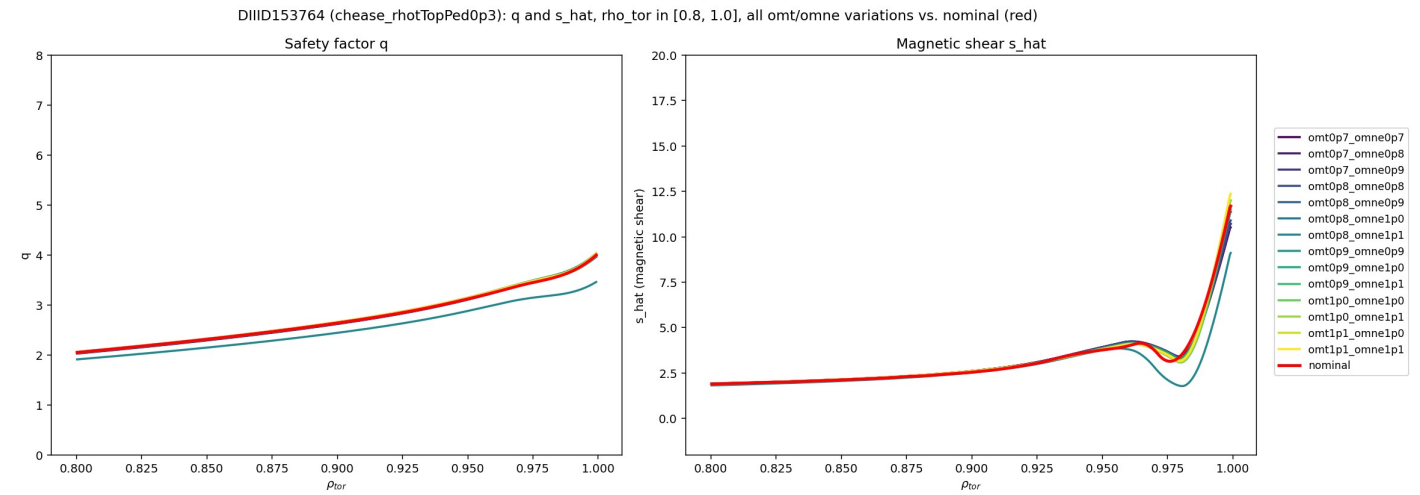
Large core profile changes (rhoTopPed = 0.3)

DIID153764

NCSCAL =4

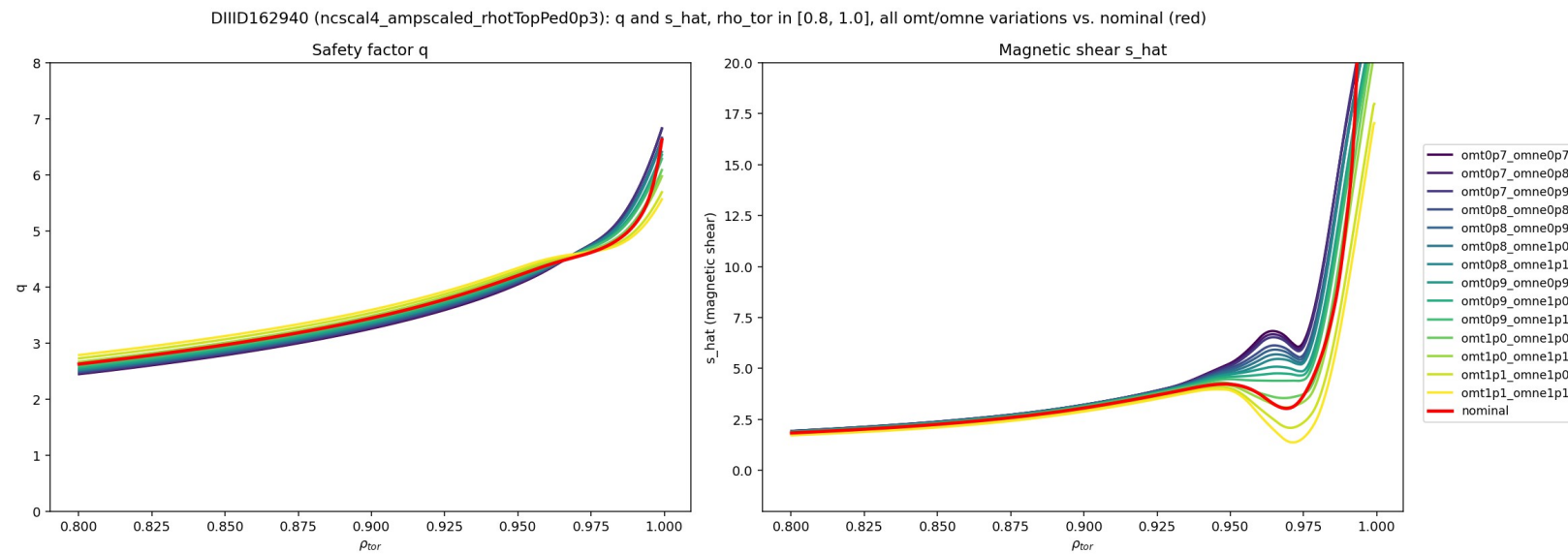


NCSCAL =1

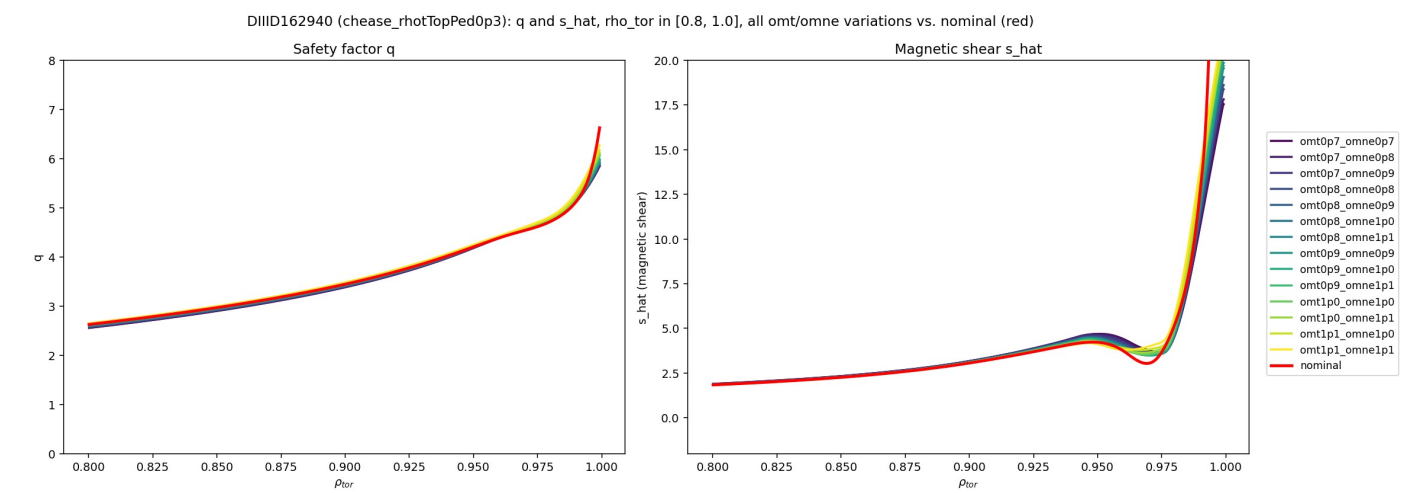


DIID162940

NCSCAL =4



NCSCAL =1



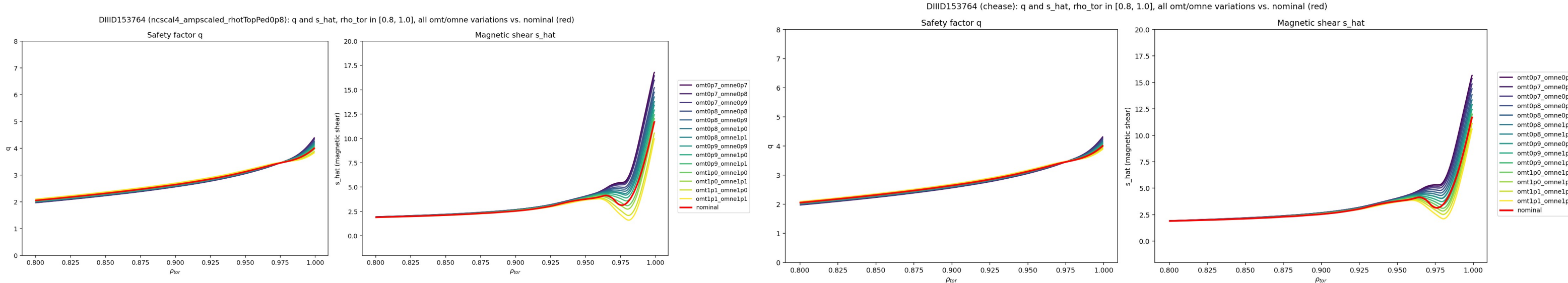
→ For both cases NCSCAL = 4 enables well converging simulations with expected shear(pressure) dependence compared to NCSCAL =1 when core profile changes are large (rhoTopPed = 0.3)

Small core profile changes (rhoTopPed = 0.8)

DIID153764

NCSCAL =4

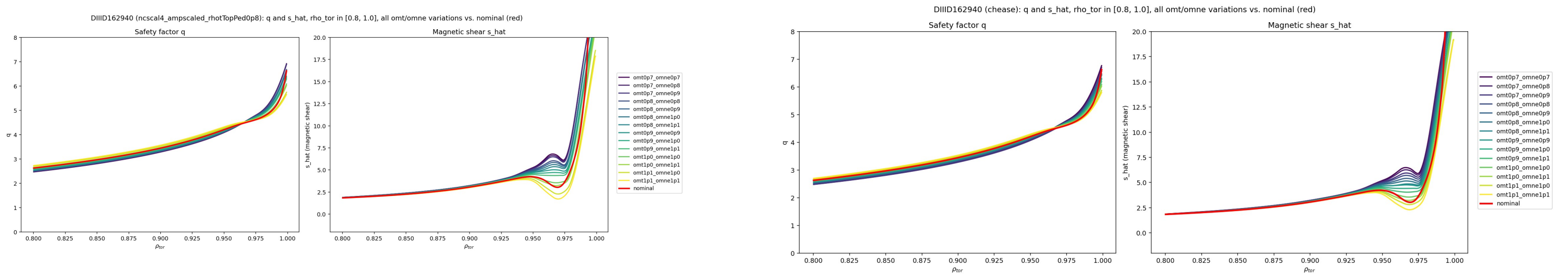
NCSCAL =1



DIID162940

NCSCAL =4

NCSCAL =1



→ For both cases NCSCAL = 4 produces q and shear profiles very close to NCSCAL =1 (fixed q0) when core profile changes are small (rhoTopPed = 0.8).